# MILITARY SPECIFICATION

# DEVICE, HIT AND LOCATING

## (For Torpedo Mk 46 Mod 0)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

## 1. SCOPE

1.1 This specification covers the Hit and Locating Device, BuWeps LD 287489. The Hit and Locating Device is of one class and one type and is referred to herein as "Hit and Locating Device." (See 6.1 for a description of the Hit and Locating Device.)

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.- The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116 Preservation, Methods of



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| | |
|---|---|
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U. S. Military Property |

DRAWINGS

Bureau of Naval Weapons

| | |
|---|---|
| BuWeps LD 287489 (List of Drawings) | Hit and Locating Device and Encasement Assembly (and all documents listed thereon) |
| BuWeps Dwg 1222243 | Battery (Mercury "A") |

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

PUBLICATIONS

Consolidated Classification Committee

Uniform Freight Classification Rules



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(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

National Bureau of Standards

| | |
|---|---|
| Handbook H28 | Screw-Thread Standards for Federal Services |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction sample.- Unless otherwise specified in the contract or order, the preproduction sample shall:

(a) Be manufactured using the procedures specified for the production lot.

(b) Consist of five Hit and Locating Devices.

(c) Be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements.-

3.2.1 Materials and treatments of materials and parts.- Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the Hit and Locating Device shall be in accordance with the requirements specified herein and the applicable documents listed in BuWeps LD 287489.

3.2.1.1 *Fungus-proof materials.-* Materials that are nutrients for fungi shall not be used where it is practical to avoid using such materials. Where used, and not hermetically sealed, materials that are nutrients for fungi shall be treated with a fungicidal agent acceptable to the procuring activity.



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3.2.2 Gages.- The supplier shall provide whatever gages are necessary and adequate to assure that the materials and parts of the Hit and Locating Device will meet dimensions and requirements shown on the applicable documents listed in BuWeps LD 287489. When requested by the supplier, the procuring activity will furnish existing pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the satisfactory fulfillment of the contract requirements.

3.2.3 Inspection and test equipment and facilities.- The supplier shall furnish and maintain all necessary inspection and test equipment and facilities, and provide personnel for performing all acceptance tests specified herein. The test equipment shall be adequate in quantity to permit performance of the required acceptance tests for the Hit and Locating Device.

3.2.4 Threads.- Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28.

3.3 Performance requirements and product characteristics.-

3.3.1 Preproduction other than acceptance requirements.-

3.3.1.1 Operational service life.- The output signal shall not decay more than three decibels (db) after three weeks of continuous, unattended, submerged operation. The current flow between the exposed connector pins and the case, when the Hit and Locating Device is submerged in sea water (3.1 percent minimum salinity), will not decrease the three-week operating life of the Hit and Locating Device, whether or not the exposed connector pins are immersed in sea water having a temperature from 29 $\pm$ 1 degree Fahrenheit (F) to 99 $\pm$ 1 degree F at any pressure up to 1000 pounds per square inch gage (psig).

3.3.1.2 Environmental requirements.- Unless otherwise specified, the Hit and Locating Device shall be capable of conforming to the performance requirements and product characteristics of this specification after being exposed, at any temperature in the applicable range, to the environments and all conditions specified in this section.



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3.3.1.2.1 Vibration (see 6.3.1).- The Hit and Locating Device shall be capable of withstanding the vibration shown in table I.

Table I

Vibration Schedule

| Condition (6.3.2) | Double Amplitude (in) minimum | Acceleration (g) minimum | Frequency (cps) |
|---|---|---|---|
| Nonoperating, Ready Non-operating and Ready | 0.75 | — | 5-6 |
| | — | 1.3 | 6-16 |
| | 0.10 | — | 16-24 |
| | — | 3.0 | 24-40 |
| | 0.036 | — | 40-74 |
| | — | 10.0 | 74-500 |

3.3.1.2.2 Shock (see 6.3.3).- The Hit and Locating Device shall be capable of withstanding shock specified in table II.

Table II

Shock Schedule

| Condition (see 6.3.2) | Peak Magnitude (g's) | Minimum Duration (ms) | Acceleration Direction |
|---|---|---|---|
| Nonoperating Ready Non-operating | 60 | 8.0 | Any |
| Ready | 45 | 20.0 | Forward |
| | 85 | 0.5 | Any except aft |
| | 300 | 0.5 | Aft |

3.3.1.2.3 Acceleration (see 6.3.4).- The Hit and Locating Device shall be capable of withstanding the accelerations specified in table III.



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Table III

Acceleration Schedule

| Condition (see 6.3.2) | Magnitude (g's) | Acceleration Direction |
|---|---|---|
| Nonoperating, Ready Non-operating and Ready | 25 | Forward |
| | 35 | Aft |

3.3.1.2.4 Temperature (see 6.3.5).- The Hit and Locating Device shall be capable of withstanding the temperatures specified in table IV.

Table IV

Temperature Schedule

| Condition (see 6.3.2) | Temperature (Degrees F) |
|---|---|
| Storage (less batteries) | -45 to +140 |
| Ready and Operating | 0 to 140 |

3.3.1.2.5 Pressure (see 6.3.6).- The fully assembled Hit and Locating Device shall operate when submerged in sea water at any pressure up to 1000 psig at any temperature from 29 ± 1 degree F to 99 ± 1 degree F for the specified operational service life (3.3.1.1).

3.3.2 Acceptance requirements.-

3.3.2.1 Activation of signal.- The Hit and Locating Device shall remain in a static ready condition until its signal is activated by one of two means:



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(a) Closing of an external circuit.

(b) The flooding of sea water (3.1 percent minimum salinity) across the exposed pins.

3.3.2.2 Output signal.- The Hit and Locating Device shall produce a pulse signal of 12 ± 2 milliseconds (ms) duration, at a repetition rate of 1.0 +0.1 -0.4 per second and with a frequency of 45 ± 1 kilocycle per second (kcps).

3.3.2.3 Output level.- The signal output level shall be 56 db referred to one dyne per square centimeter (dyne/$cm^2$) at one yard from the Exercise Head longitudinal center line.

3.3.2.4 Power supply.- The Power supply for the Hit and Locating Device shall be self-contained batteries (BuWeps Drawing 1222243) capable of an output of 17.5 ± 1.0 volt direct current (vdc).

3.3.2.5 Attitude.- The Hit and Locating Device shall be capable of operating in accordance with the performance requirements when positioned in any attitude.

3.3.2.6 Weight.- The weight of the Hit and Locating Device, less mounting hardware, shall not exceed five pounds.

3.3.2.7 Workmanship.- The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

3.3.2.8 Identification.- The Hit and Locating Device shall be permanently identified in accordance with Standard MIL-STD-130 and the applicable documents listed in BuWeps LD 287489.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Reliability tests.-

4.2.1 Classification of reliability tests.- The classification of reliability tests and the tests and examinations required under each classification are as follows:

(a) Preproduction tests other than acceptance tests (4.7).

(1) Operational service life test (4.7.1)

(2) Environmental tests (4.7.2)

(b) Acceptance tests and examination (4.8).

(1) Operational performance test (4.8.1.2)

(2) Examination (4.8.1.3)

4.3 Sampling.- Unless otherwise specified, and when applicable, the sampling plans and procedures used in determining the acceptability of products shall be in accordance with the provisions set forth in Standard MIL-STD-105.

4.3.1 Preproduction sampling.- Unless otherwise specified, for each contract or order, a lot of five Hit and Locating Devices shall be manufactured using the facilities, methods, and procedures proposed for production. The preproduction sample shall be subjected to preproduction tests at an activity designated in the contract or by the Bureau of Naval Weapons for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the Hit and Locating Devices by the supplier, prior to approval of the preproduction



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sample, shall be at the supplier's risk. Preproduction samples accepted will be applied as part of the quantity specified in the contract or order, but shall not be used in any equipment manufactured for the Government.

4.3.2 Production sampling.- Inspection lot definition, formation, and size shall be in accordance with Standard MIL-STD-105.

4.3.3 Periodic production sampling.- Unless otherwise specified, at any time during production of the Hit and Locating Devices for the contract or order, a random sample of two Hit and Locating Devices will be selected for submission to a Bureau of Naval Weapons assigned activity for the periodic production tests. The Hit and Locating Devices shall be considered satisfactory if they meet all the requirements of the applicable contract, specifications, and drawings and withstand the tests specified herein. Periodic production samples tested will be applied as part of the quantity specified by the contract or order but shall not be used in any equipment manufactured for the Government.

4.4 Rejection.-

4.4.1 Rejection of preproduction samples.- Failure of any Hit and Locating Device in the preproduction sample to meet the requirements specified herein, after being tested in accordance with the preproduction tests specified in this section, shall be cause for rejection of all the Hit and Locating Devices in the preproduction sample.

4.4.2 Rejection of production samples.- Failure of any Hit and Locating Device in the production sample (inspection lot, see 4.3.2) to meet the acceptance requirements, after being tested in accordance with the acceptance tests set forth in this section, may be considered cause for rejection of the entire production sample (inspection lot) or cause for cessation of production as determined by the procuring activity.

4.5 Test equipment.- The following items of the test equipment, or equivalent, will be required to perform the acceptance tests specified herein.

(a) Voltmeter - Capable of measuring 0 to 20 vdc with a minimum accuracy of $\pm$ 0.2 vdc over full scale deflection.



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(b) Hewlett-Packard Universal Counter Model 523D

(c) Tektronix Oscilloscope Model 535A

(d) Hewlett-Packard Power Supply Model 723A

(e) Tektronix Probe P6000

(f) Tektronix Probe P6027

(g) Stop Watch - With an accuracy of one-half percent in one minute.

(h) Test chamber - The chamber shall be a temperature controlled pressure chamber capable of the following:

(1) Sensing the output of the Hit and Locating Device.

(2) Being pressurized to 1000 psig using salt water (common salt, sodium chloride in tap water, specific gravity 1.025 ± 0.015) and must be able to retain the pressure of 1000 psig for a period of three weeks.

(3) Maintaining salt water at a temperature of 29 ± 1 degree F for a period of two weeks and 99 ± 1 degree F for a period of one week.

4.6 Test conditions.- Unless otherwise specified, the Hit and Locating Device shall be subjected to the tests under the following conditions:

(a) Temperature - Room ambient (65 to 95 degrees F)

(b) Altitude - Normal ground atmospheric pressure

(c) Vibration - None

(d) Humidity - (relative) - Room ambient to 95 percent

(e) Attitude - Any

4.7 Preproduction tests, other than acceptance tests.- After the following tests (4.7.1 through 4.7.2.5), each Hit and Locating Device shall be subjected to, and satisfactorily pass, the acceptance tests and examination (4.8).



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4.7.1 Operational service life test.- The fully assembled Hit and Locating Device shall be tested under conditions which simulate service use. The operational service life test shall determine proper operation and duration of the service life of the Hit and Locating Device. The Hit and Locating Device shall be tested for 2 weeks in the dry state, that is, with the exposed connector pins not immersed in the salt water. The Hit and Locating Device shall be tested for one week in the wet state, that is, with the two exposed connector pins immersed in salt water (3.1 percent salinity).

(a) Place the Hit and Locating Device in the test chamber (4.5 (h)).

(b) Fill the test chamber with salt water cooled to a temperature of 29 ± 1 degree F.

(c) Pressurize the test chamber to 1000 ± 10 psig.

(d) Maintain salt water temperature of 29 ± 1 degree F and maintain chamber pressure at 1000 ± 10 psig for two weeks.

(e) With the two exposed connector pins dry (not immersed in salt water) activate the output signal of the Hit and Locating Device with the metallic connection. Monitor the output signal to verify compliance with the requirements of 3.3.2.1, 3.3.2.2 and 3.3.2.3.

(f) Raise temperature of salt water to 99 ± 1 degree F.

(g) Maintain chamber pressure of 1000 psig and salt water temperature at 99 ± 1 degree F for a period of one week.

(h) With the two exposed connector pins immersed in salt water (3.1 percent salinity) monitor output signal of the Hit and Locating Device to verify compliance with requirements of 3.3.2.1, 3.3.2.2 and 3.3.2.3 and 3.3.1.1.



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4.7.2 Environmental tests.- The Hit and Locating Device shall be subjected to the following environmental tests, and thereafter shall meet the performance requirements and product characteristics of this specification.

4.7.2.1 Vibration.- Mount the Hit and Locating Device as shown in figure 1. The Hit and Locating Device shall be conditioned at room temperature for a minimum period of one hour prior to conducting each resonance survey. Vibrate the unit for a total of 9 hours (see 3.3.1.2.1, and table I) in accordance with the requirements of Procedure XII, Specification MIL-E-5272 and the vibration deviations of 4.7.2.1.1.

4.7.2.1.1 Vibration deviations.- The following are deviations from Procedure XII, Specification MIL-E-5272. Individual resonance surveys shall be conducted with vibration applied along three mutually perpendicular axes. The resonant and cycling period in each axis shall be three hours, at room temperature, in accordance with table V.

Table V

Resonant and Cycling Schedule

| No. of Resonances | Total Time at Resonances (Minutes)* | Cycling Time (Minutes) |
|---|---|---|
| 0 | 0 | 180 |
| 1 | 27 | 153 |
| 2 | 54 | 126 |
| 3 | 81 | 99 |
| 4 | 108 | 72 |

*Twenty-seven minutes at each resonance.



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4.7.2.2 Shock (see 6.3.3).- Mount the Hit and Locating Device as shown in figure 1 and secure it to a shock test fixture, at room temperature, so that a minimum of 90 percent of the table input to the fixture will be applied to the Hit and Locating Device under test. Subject the Hit and Locating Device to the shock listed in table VI as specified in the Shock Test, Procedure IV, Specification MIL-E-5272 except as follows. Shock magnitude and duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock. Test equipment is not restricted to that specified in Specification MIL-E-5272.

Table VI

Shock Schedule

| Condition (see 6.3.2) | Peak Magnitude (g's) | Duration (ms) | Impact Direction | Total No. of Impacts |
|---|---|---|---|---|
| Ready | 300 | 0.5 to 1.0 | Longitudinal, Aft | 3 |
| | 85 | 0.5 to 1.0 | Any except Longitudinal Aft | 6 |
| Nonoperating | 60 | 8.0 to 8.5 | Any | 9 |
| Ready | 45 | 20.0 to 21.0 | Longitudinal Forward | 3 |

4.7.2.3 Acceleration.- The Hit and Locating Device shall be mounted as shown in figure 1 and subjected to the acceleration magnitudes and directions required by table III for six minutes at room temperature.

4.7.2.4 Temperature, storage.- The Hit and Locating Device, preserved and packaged in accordance with 5.1.1, less batteries, shall be placed in a temperature controlled chamber and subjected to the following temperature cycle:

(a) Heat to 140 +5 -0 degrees F within 15 minutes

(b) Maintain at 140 +5 -0 degrees F for four hours



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(c) Cool to -45 +0 -5 degrees F within 15 minutes

(d) Maintain at -45 +0 -5 degrees F for four hours

(e) Return to room temperature within 15 minutes

Repeat this schedule for a minimum of three cycles.

4.7.2.5 Pressure.- The Hit and Locating Device shall be tested under test conditions set forth in 4.6. The Hit and Locating Device shall meet the pressure requirements of 3.3.1.2.5.

4.8 Acceptance tests and examination.- Unless otherwise specified in the contract or order, the supplier shall be responsible for conducting the acceptance tests on each Hit and Locating Device to be delivered. The supplier shall maintain adequate inspection records acceptable to the procuring activity. The acceptance tests and examination shall be as follows:

(a) Operational performance test (4.8.1.2)

(b) Examination of product (4.8.1.3)

4.8.1 Acceptance test setup and procedure.-

4.8.1.1 Operational test setup (see figure 2, test connections).-

(a) Remove the case housing from the Hit and Locating Device.

(b) Disconnect the battery.

(c) Connect the power supply (4.5 (d)) to the battery input terminals.

(d) Connect the oscilloscope (4.5 (c)) with P6000 probe (4.5 (e)) to the primary of the output transformer.

(e) Adjust the oscilloscope for 10 volts per centimeter vertical deflection sensitivity input level and two ms per centimeter horizontal sweep rate.



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(f) Connect counter (4.5 (b)) with probe P6027 (4.5 (f)) to vertical output of oscilloscope.

(g) Adjust counter(4.5 (b)) for a 10-period average measurement.

4.8.1.2 Operational performance test.- With the Hit and Locating Device connected to test equipment as in 4.8.1.1, conduct the following test and record results as follows:

(a) Adjust power supply (4.5 (d)) to supply an output of 17.5 ± 1.0 vdc.

(b) Using the stop watch (4.5 (g)), count pulses for a 60-second period. The number of pulses counted shall be between 55 and 100 pulses to verify repetition rate of 1.0 +0.1 -0.4 per second (3.3.2.2).

(c) Observe peak-to-peak voltage of pulses on the oscilloscope (4.5 (c)). The peak-to-peak voltage shall be greater than 350 volts to verify the output level of 56 db (3.3.2.3).

(d) Depress reset on counter (4.5 (b)) and then release immediately after a pulse and then note the time after the next pulse. Time recorded shall be between 21.7 and 22.7 microseconds to verify the frequency of 45 ± 1 kcps (3.3.2.2).

(e) Change the input connection to the counter to EXTERNAL and set counter function to MANUAL GATE. Open manual gate switch and reset counter. Close manual gate switch after 10 pulses. Record count. Count shall be between 4400 and 6440 to verify pulse duration of 12 ± 2 ms (3.3.2.2).

4.8.1.3 Examination of product.- The Hit and Locating Device shall be examined for conformance to the requirements of this specification and BuWeps LD 287489. The examination shall be conducted as specified in table VII and any Hit and Locating Device in the sample, containing one or more defects, shall be rejected. If the number of defective Hit and Locating Devices in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.



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Table VII

Classification of defects in accordance with Standard MIL-STD-105.

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | 1.0% AQL |
| 101 | Component parts missing |
| 102 | Leads not soldered |
| 103 | Adhesive not applied to threads |
| 104 | Visible holes, cracks or perforation in housing |
| 105 | Component parts not new; evidence of reclaimed, used or salvaged components |
| 106 | Unit configuration extends beyond drawing limits; dimensions do not comply with drawing requirements |
| 107 | Evidence of unauthorized material and repairs |
| 108 | Battery terminals not properly identified; polarity of terminals incorrect; output of battery less than 17.5 $\pm$ 1.0 vdc (3.3.2.4). |
| Minor: | 2.5% AQL |
| 201 | Threads not in accordance with drawing requirements and Handbook H28 |
| 202 | Unit weight above drawing tolerance |
| 203 | Identification and markings not located as specified; not legible or data incomplete |
| 204 | Evidence of damage to housing; scratches or small dents |
| 205 | Preservation, packing, or packaging of the Hit and Locating Device is not as specified in section 5 of this specification. |

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.4).



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5.1.1 Level A.- Cleaning, drying, preservation, and packaging methods for level A shipment shall be in accordance with Specification MIL-P-116 Process C-1 and Method IA-8.

5.1.2 Level B.- Not applicable.

5.1.3 Level C.- Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been proven by testing to conform to all applicable requirements.

5.2 Packing.-

5.2.1 Level A.- The Hit and Locating Device, preserved and packaged in accordance with 5.1.1, shall be packed in a container in accordance with Specification MIL-P-116, Method IA-2.

5.2.2 Level B.- Not applicable.

5.2.3 Level C.- The Hit and Locating Device, packaged as specified in 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or common carrier regulations applicable to the mode of transportation.

5.3 Marking.-

5.3.1 Normal marking.- In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Description and intended use.-

6.1.1 Description.- The Hit and Locating Device is a self-contained unit which can function independent of any other system. The Hit and Locating Device is flange mounted and has a transducer mounted to the exposed curved surface. The transducer transmits a pulsed signal through the exposed surface. The Hit and Locating Device remains in a static,



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ready condition, until actuation is effected by either a metallic connection or by sea water (3.1 percent salinity) flooding across the pins. The output provides a continuous signal for a minimum period of three weeks. The sealed case of the Hit and Locating Device will withstand hydrostatic pressures up to 1000 psig.

6.1.2 Intended use.- The Hit and Locating Device is intended for use in the Exercise Head of the Torpedo Mark 46 Mod 0 as a signaling device to aid in locating the torpedo after a practice run.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Required level of preservation and packaging.

(c) Assigned activity for preproduction and periodic tests.

(d) Any exceptions to this specification.

6.3 Definitions.-

6.3.1 Vibration.- Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.2 Hit and Locating Device condition.-

6.3.2.1 Nonoperating.- The condition of the uninstalled Hit and Locating Device when stored, transported, or handled.

6.3.2.2 Ready, nonoperating.- The condition of the Hit and Locating Device when assembled with the end item, transported and handled.

6.3.2.3 Ready.- The condition of the Hit and Locating Device when fully assembled with the end item and installed on the launching mechanism prior to operating.



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6.3.2.4 Operating.- Operating is the condition of the Hit and Locating Device when operating.

6.3.3 Shock.- Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), duration in units of time (milliseconds (ms)), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

6.3.4 Acceleration (steady state).- Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.3.5 Temperature.- Temperature is the temperature of the environment to which the Hit and Locating Device is exposed as expressed in degrees Fahrenheit.

6.3.6 Pressure.- Unless otherwise specified, pressure is the pressure of the environment to which the Hit and Locating Device is exposed, expressed in pounds per square inch absolute (psia) or pounds per square inch gage (psig).

6.4 Preparation for delivery criteria.- Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A.- This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level B.- Not applicable.

Level C.- This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.



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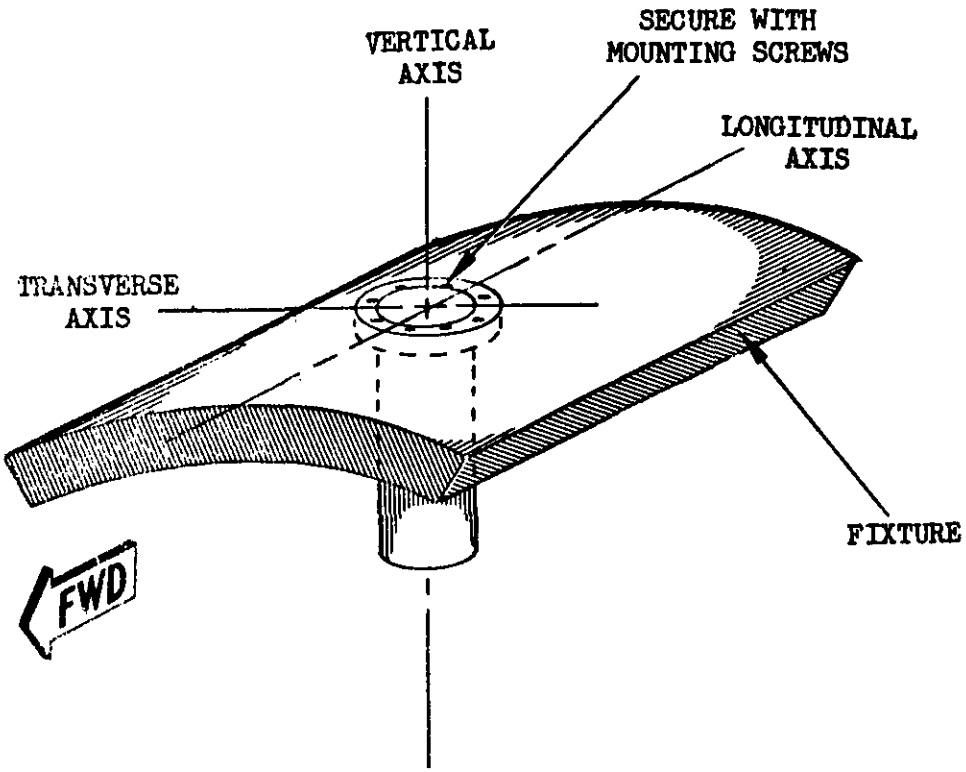



FIGURE -1- AXES OF THE HIT AND LOCATING DEVICE



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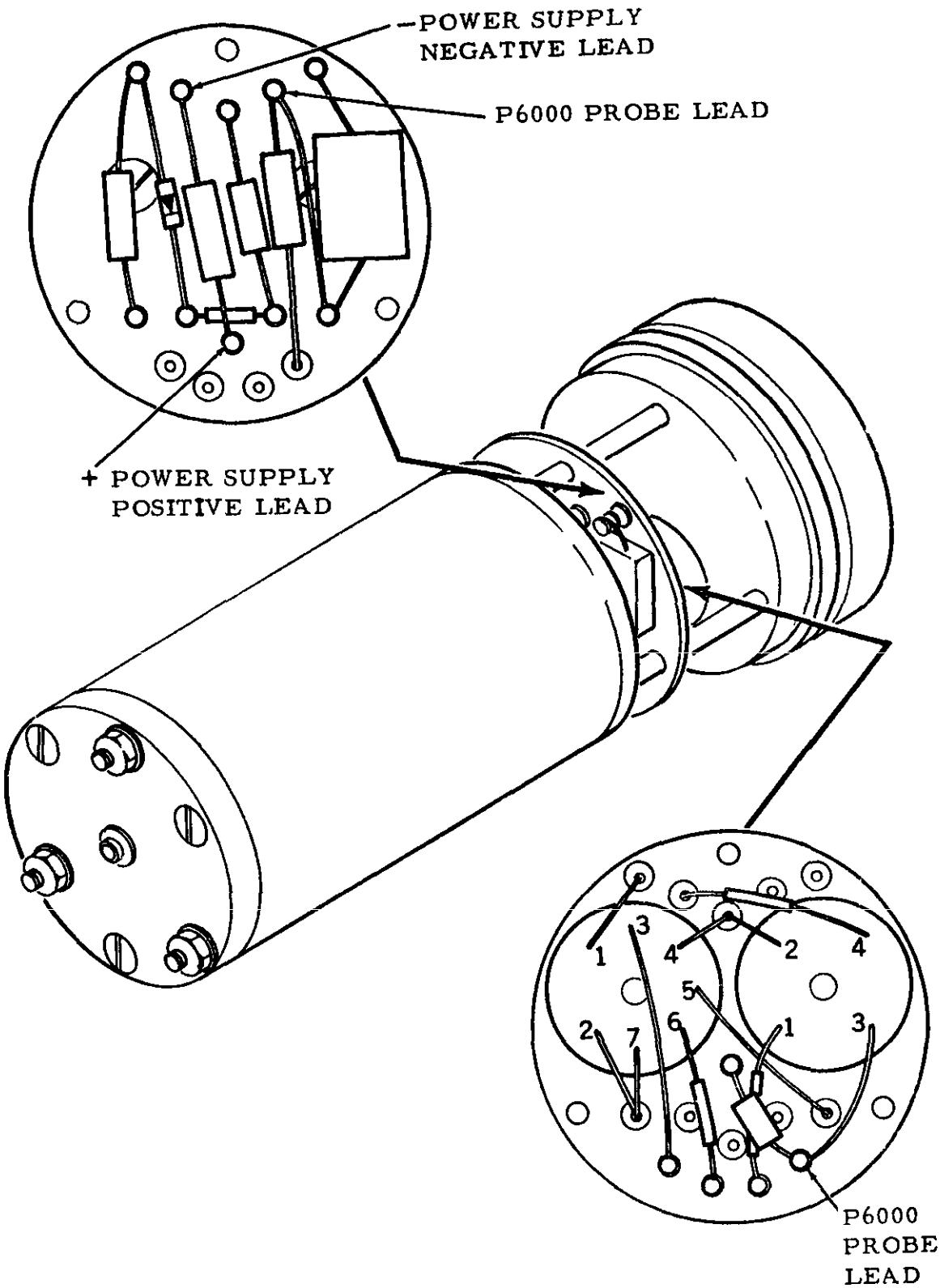



FIGURE -2- TEST CONNECTIONS



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-D-81068(Wep) DEVICE, HIT AND LOCATING (FOR TORPEDO MK 46 MOD 0)

| ORGANIZATION *(Of submitter)* | CITY AND STATE |
|---|---|
| | |

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

BUREAU OF NAVAL WEAPONS (RREN-5)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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